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Key Points:

- An efficient and accurate pattern synthesis method for radome-enclosed planar antenna arrays in radar applications is presented
- Iterative fast Fourier transform via two-dimensional least-square active element pattern expansion is proposed
- A blocked accumulating matrix is introduced which relates the virtual and physical excitation vectors

Correspondence to:

Y. Liu,
yhliu@uestc.edu.cn

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Efficient and Accurate Pattern Synthesis for Radome-Enclosed Planar Arrays Using Iterative FFT via Two-Dimensional Least-Square Active Element Pattern Expansion

Jingjing Bai¹, Yanhui Liu^{1,2} , Shiwen Yang², Jun Hu², and Zaiping Nie²

¹Yangtze Delta Region Institute (Quzhou), University of Electronic Science and Technology of China (UESTC), Quzhou, China, ²School of Electronic Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China

Abstract This work generalizes the iterative fast Fourier transform (FFT) via least-square active element pattern expansion method to efficiently synthesize two-dimensional (2D) patterns with accurate sidelobe and mainlobe shape control for planar arrays with radomes. The 2D active element patterns (2D-AEPs) of a planar array-radome structure are utilized to include the mutual coupling and radome effect. The 2D-LSAEPE method is presented to approximate each 2D-AEP as the pattern by exciting a virtual rectangular subarray with identical element patterns, and the weighting coefficients of the virtual subarray are found by minimizing the least-square error of the 2D-AEP approximation. With the help of the 2D-LSAEPE, the pattern of a planar array with a radome can be efficiently calculated by FFT. To apply the FFT via 2D-LSAEPE for planar array pattern computation, a blocked accumulating matrix is introduced which relates the virtual and physical excitation vectors. In addition, an interpolation-based pre-processing technique is implemented to obtain the virtual element pattern data in uniform grid of the (u, v) -space so as to meet the requirement of employing the 2D-FFT. By integrating the FFT with the 2D-LSAEPE and the interpolation technique, an efficient iterative method is developed to implement accurate pattern synthesis for planar arrays with radomes. Three examples of synthesizing different planar array-radome systems are conducted. Synthesis results show that the proposed method achieves much more accurate synthesis result than the traditional FFT-based technique, while taking much less computation time than some other methods. This verifies the effectiveness of the proposed method.

1. Introduction

Antenna systems are of great importance for communication, sensing, radar systems, and etc. For example, in wireless communications, multiple input multiple output antenna systems have been widely studied to meet the requirements of the fifth generation mobile communication (Alibakhshikenari, Virdee, Benetatos, et al., 2022), and some metamaterial-based antenna systems are attractive for further millimeter-wave (mm-wave) and terahertz (THz) wireless communications (Alibakhshikenari, Ali, et al., 2022; Alibakhshikenari, Virdee, Althuwayb, et al., 2021). In radar systems, phased arrays are widely used because of their pattern mainlobe shape and sidelobe level (SLL) control capability, as well as some other superior properties (Fenn et al., 2000; Kouemou, 2010). However, it is known that antenna arrays are usually used together with radomes which protect antenna arrays from the influence of outside environments in practice. The presence of the radome usually degrades the pattern performance of the antenna array (Frech et al., 2013; L. Liu & Nie, 2019). Thus, to suppress the radome effect, a number of techniques have been presented for optimizing radome's parameters such as its shape and thickness distribution (Hsu et al., 1993; Xu et al., 2017). However, the freedom of radome optimization is sometimes very limited since the radome should also meet other requirements such as its mechanical strength and electromagnetic scattering characteristics. Compared with the modification of a radome, optimizing element excitation amplitudes and phases by considering the antenna array-radome as a whole system would be a more convenient alternative to compensate for the radome effect if a phased antenna array is considered.

In the past, many array pattern synthesis methods have been presented, but most of them deal only with the array factor synthesis, ignoring the mutual coupling (Ares-Pena et al., 1999; Capozzoli et al., 2022; Caratelli & Viganó, 2011; D'Urso et al., 2016; Poli et al., 2010). To suppress the mutual coupling between elements, some techniques of adding appropriate isolators around antenna elements have been presented (Alibakhshikenari, Virdee, Benetatos, et al., 2022; Alibakhshikenari, Virdee, Salekzamankhani, et al., 2021; Alibakhshikenari, Virdee, Vadalà, et al., 2022). Additionally, in phased arrays, some methods using active

element patterns (AEPs) such as in Kelley and Stutzman (1993), Pirhadi et al. (2014), and Pozar (1994) can take into account the element mutual coupling of antenna arrays in pattern synthesis, but they usually do not consider the radome effect. Nevertheless, there exist several techniques which have incorporated the radome effect into the pattern synthesis. In Landesa et al. (1999), a direct solving technique based on method of moments is proposed to optimize pattern performance for arrays close to dielectric bodies, but this technique still used the ideal point source assumption without considering the antenna element structure. In Dai et al. (2021) and Zhou et al. (2020), the concept of AEPs is extended where all the AEPs are obtained by full-wave simulating the practical antenna array-radome structure to include antenna element mutual coupling as well as radome effect. In these techniques, stochastic algorithms such as simulated annealing in Zhou et al. (2020), genetic algorithm and particle swarm optimization (PSO) in Dai et al. (2021) are applied to find the optimized excitation amplitudes and phases for improved pattern performance. Despite their success, these stochastic algorithms are time consuming since a large number of array pattern calculations based on direct summation of all the different AEPs are used. Other methods such as convex optimization (Frid & Jonsson, 2018; Fuchs, 2012) and semidefinite relaxation (SDR) method (Y. Liu et al., 2018), can be also extended to include the radome effect and mutual coupling into the pattern synthesis by utilizing the AEPs of a practical array-radome system. However, these methods are still inefficient for planar antenna array case because a huge number of inequality constraints should be used for two-dimensional (2D) pattern shape control. Up to now, efficiently synthesizing the array pattern considering radome effect remains a very challenging problem.

It is well known that the Fourier transform (FT) relationship exists between the excitation distribution and the array factor of a uniformly spaced linear or planar array, and it can be utilized to accelerate the pattern computation (Chou et al., 2004). Based on this relationship, an iterative FT technique (IFT) was developed where the excitations of an evenly spaced array can be efficiently obtained through iteratively employing the fast Fourier transform (FFT) and inverse FFT (IFFT) in each iteration (Keizer, 2007, 2009b). The IFT method was then extended to synthesize multiple-pattern planar arrays (Li et al., 2016) and thinned planar arrays (Keizer, 2009a; Wang et al., 2014). By replacing the FFT with nonuniform FFT (NUFFT), the IFT method can be modified to be suitable for unequally spaced array synthesis (Yang et al., 2013). Thanks to the use of FFT or NUFFT, these methods are highly computationally efficient. However, they are only applicable for synthesizing array factors without considering mutual coupling of practical antenna arrays. Recently, an iterative FFT via least-square active element pattern expansion (LSAEPE) method was presented to tackle the efficient synthesis of linear arrays including coupling and platform effects (Bai et al., 2021; Huang et al., 2017). In this method, each AEP of an equally spaced linear array is regarded as the radiated pattern from several neighboring virtual elements having identical element patterns, so that the array pattern including mutual coupling and platform effect can be efficiently computed. Nevertheless, for a planar array including mutual coupling and radome effect, applying the FFT-based technique to achieve accurate and efficient pattern synthesis has never been discussed in literature.

In this work, the iterative FFT via LSAEPE method is generalized with significant modification to efficiently synthesize two-dimensional (2D) patterns of planar arrays enclosed with radomes. A 2D-LSAEPE is developed in which 2D-AEPs of the whole planar array-radome system are adopted to include the radome effect and mutual coupling, and each AEP is regarded as the pattern radiated by a nearby virtual rectangular subarray. Thus the 2D-AEP can be mathematically approximated by summing several phase-shifted virtual element patterns multiplied with appropriate coupling coefficients. Since the 2D-AEP obtained for the array-radome system is an inseparable function, the optimal coupling coefficients in terms of the least-square error cannot be simply expressed by using product of two one dimensional (1D) coefficient vectors. To integrate the FFT with 2D-LSAEPE for planar array case, a blocked accumulating matrix is introduced which relates the virtual and physical excitation vectors by accumulating multiple coupling coefficients from different virtual subarray. Besides, the virtual element pattern in the 2D-LSAEPE is obtained by averaging over all the AEPs that are usually sampled in the (θ, φ) -space. Such pattern sampling grid does not conform with the 2D-FFT requirement. Thus, an interpolation-based pre-processing technique is presented to obtain the samples of the virtual element pattern in the uniform grid of (u, v) -space. By integrating the FFT with the proposed 2D-LSAEPE, an iterative method is developed to achieve accurate and efficient pattern synthesis for planar arrays enclosed with radomes. Three examples of synthesizing pencil beam, shaped beam and scannable beam for different planar antenna arrays in presence of different types of radomes are provided to verify the efficiency and accuracy of the proposed method. The performance comparisons with other methods are also provided in the examples.

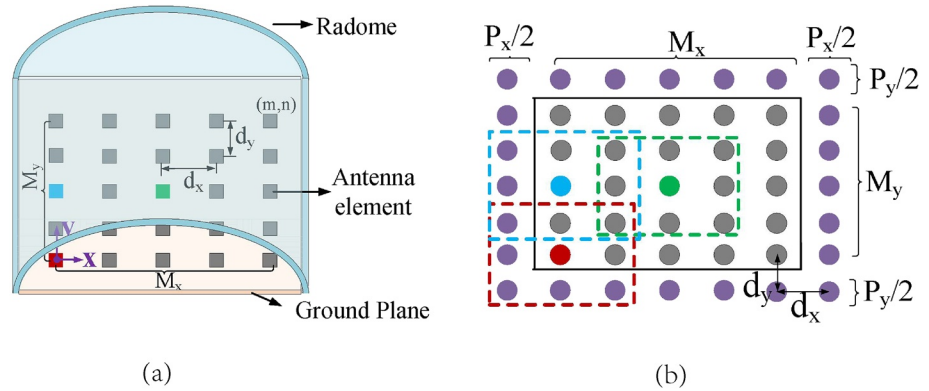


Figure 1. (a) A real planar array-radome system composed of $M_x \times M_y$ elements, and (b) An enlarged virtual planar array consisting of $(M_x + P_x) \times (M_y + P_y)$ virtual elements. The area framed by the red, green, and blue dotted lines in the virtual array is utilized to approximate the active element pattern of the element indicated by the corresponding color block in the real array, respectively, via the proposed 2D-LSAEPE method.

2. Formulation and Algorithm

2.1. Pattern Synthesis of a Planar Array Enclosed With a Radome

Consider a rectangular planar array placed in a radome, and the array is composed of $M_x \times M_y$ elements distributed along rectangular grids with uniform spacing d_x and d_y in the x - and y -directions, respectively. The geometry of this array-radome system is shown in Figure 1a. Then, its $\vec{e}_\theta$ - and $\vec{e}_\varphi$ -polarized far-field pattern can be expressed as

$$\begin{bmatrix} F_\theta(\theta, \varphi) \\ F_\varphi(\theta, \varphi) \end{bmatrix} = \sum_{m=0}^{M_x-1} \sum_{n=0}^{M_y-1} w_{m,n} e^{j\beta(m d_x u + n d_y v)} \begin{bmatrix} g_\theta^{(m,n)}(\theta, \varphi) \\ g_\varphi^{(m,n)}(\theta, \varphi) \end{bmatrix} \quad (1)$$

where $w_{m,n}$ is the complex excitation of the (m, n) th element, $\beta = 2\pi/\lambda$ with λ is the wavelength, $u = \sin\theta\cos\varphi$ and $v = \sin\theta\sin\varphi$. $g_\theta^{(m,n)}(\theta, \varphi)$ and $g_\varphi^{(m,n)}(\theta, \varphi)$ are the $\vec{e}_\theta$ - and $\vec{e}_\varphi$ -polarized components of the 2D-AEP of the (m, n) th antenna element in the planar array-radome system. Note that the 2D-AEPs are utilized here to take the mutual coupling and radome effect into account (Pozar, 1994).

In practice, we expect the array pattern in the desired polarization direction to satisfy a specified spatial distribution shape. Usually, the user-defined polarization direction indicated by $\vec{p}_d$ is a known fixed direction, and it depends on the practical application. Through projecting $\vec{e}_\theta$ - and $\vec{e}_\varphi$ -polarized components of the (m, n) th 2D-AEP to the desired polarization direction $\vec{p}_d$, the desired element pattern of the (m, n) th antenna can be obtained by

$$g^{(m,n)}(\theta, \varphi) = [\vec{e}_\theta g_\theta^{(m,n)}(\theta, \varphi) + \vec{e}_\varphi g_\varphi^{(m,n)}(\theta, \varphi)] \cdot \vec{p}_d \quad (2)$$

Then, the desired array pattern for the planar antenna array in presence of the radome can be expressed as

$$F(\theta, \varphi) = \sum_{m=0}^{M_x-1} \sum_{n=0}^{M_y-1} w_{m,n} g^{(m,n)}(\theta, \varphi) e^{j\beta(m d_x u + n d_y v)} \quad (3)$$

2.2. Two-Dimensional Active Element Pattern Expansion Method

Due to the mutual coupling and radome effect, the 2D-AEPs obtained for the array-radome system generally differ from each other. Therefore, the traditional 2D-FFT cannot be directly applied to speed up the pattern computation in Equation 3. To solve this issue, we extend the idea of LSAEPE addressed in Huang et al. (2017) to handle the 2D-AEPs expansion for planar arrays with radomes, and this extension is called two-dimensional LSAEPE (2D-LSAEPE). In this extended method, each 2D-AEP is approximated as the pattern radiated by a virtual planar subarray composed of nearby $(P_x + 1) \times (P_y + 1)$ elements with identical element patterns, as

shown in Figure 1. The identical element pattern of the virtual subarray is mathematically found by averaging over all 2D-AEPs. That is, finding the following approximation

$$g^{(m,n)}(\theta, \varphi) \approx g^{\text{ave}}(\theta, \varphi) \sum_{p=-\frac{P_x}{2}}^{\frac{P_x}{2}} \sum_{q=-\frac{P_y}{2}}^{\frac{P_y}{2}} c_{p,q}^{(m,n)} e^{j\beta(pd_x u + qd_y v)} \quad (4)$$

where $g^{\text{ave}}(\theta, \varphi) = \sum_{m=0}^{M_x-1} \sum_{n=0}^{M_y-1} g^{(m,n)}(\theta, \varphi) / (M_x \times M_y)$ presents the averaged element pattern. Mathematically, each 2D-AEP is approximated as weighted summation of phase-shifted averaged element patterns. To approximate the 2D-AEPs for the edge elements, $P_x/2$ columns and $P_y/2$ rows of virtual elements are added into each side of the original array, which are indicated by purple solid circles as shown in Figure 1b. This obtained new array is called as the enlarged virtual planar array which is composed of $(M_x + P_x) \times (M_y + P_y)$ elements. It should be noted that the virtual array only exists mathematically, but does not really exist in practice. $c_{p,q}^{(m,n)}$ means the coupling coefficient of the $(m + p, n + q)$ th element with respect to the (m, n) th one, and it can be calculated by minimizing the least-square (LS) error of the AEP approximation given by

$$\min_{\mathbf{c}^{(m,n)}} \|\mathbf{Z} \cdot \mathbf{\tilde{c}}^{(m,n)} - \mathbf{g}^{(m,n)}\|_2^2 \quad (5)$$

where

$$\mathbf{Z} = \mathbf{G} \odot \tilde{\mathbf{V}} \quad (6)$$

and

$$\mathbf{G} = \begin{bmatrix} g^{\text{ave}}(\theta_1, \varphi_1) & & & O \\ & g^{\text{ave}}(\theta_2, \varphi_2) & & \\ & & \ddots & \\ O & & & g^{\text{ave}}(\theta_J, \varphi_J) \end{bmatrix} \quad (7)$$

$$\tilde{\mathbf{V}} = \begin{bmatrix} \mathbf{V}_{-\frac{P_y}{2}}, & \mathbf{V}_{-\frac{P_y}{2}+1}, & \dots, & \mathbf{V}_{\frac{P_y}{2}} \end{bmatrix} \quad (8)$$

$$\mathbf{V}_q = \begin{bmatrix} e^{j\beta\left(-\frac{P_x}{2}d_x u_1 + qd_y v_1\right)} & \dots & e^{j\beta\left(\frac{P_x}{2}d_x u_1 + qd_y v_1\right)} \\ \vdots & \ddots & \vdots \\ e^{j\beta\left(-\frac{P_x}{2}d_x u_J + qd_y v_J\right)} & \dots & e^{j\beta\left(\frac{P_x}{2}d_x u_J + qd_y v_J\right)} \end{bmatrix} \quad (9)$$

$$\mathbf{g}^{(m,n)} = \left[g^{(m,n)}(\theta_1, \varphi_1), \dots, g^{(m,n)}(\theta_J, \varphi_J) \right]^T \quad (10)$$

$$\mathbf{\tilde{c}}^{(m,n)} = \left[\left\{ \mathbf{c}_{-\frac{P_y}{2}}^{(m,n)} \right\}^T, \dots, \left\{ \mathbf{c}_{\frac{P_y}{2}}^{(m,n)} \right\}^T \right]^T \quad (11)$$

$$\mathbf{c}_q^{(m,n)} = \left[c_{-\frac{P_x}{2},q}^{(m,n)}, \dots, c_{\frac{P_x}{2},q}^{(m,n)} \right]^T \quad (12)$$

In the above, the symbol “ $\odot$ ” denotes the element-wise product. The coupling coefficient vector $\mathbf{\tilde{c}}^{(m,n)}$ can be analytically obtained by $\mathbf{\tilde{c}}^{(m,n)} = (\mathbf{Z}^H \mathbf{Z})^{-1} \mathbf{Z}^H \mathbf{g}^{(m,n)}$ where $(\cdot)^H$ denotes the conjugate transpose operation of a matrix. Note that $(\mathbf{Z}^H \mathbf{Z})^{-1} \mathbf{Z}^H$ is independent of (m, n) , and consequently it only needs to be calculated once for different 2D-AEPs. Here, since the 2D-AEP obtained for the array-radome system is an inseparable function, the coupling coefficient $\mathbf{\tilde{c}}^{(m,n)}$ cannot be simply expressed by using product of two one dimensional (1D) coefficient vectors.

2.3. Fast Transformation Between Array Pattern and Excitation Distribution Using IFFT/FFT via 2D-LSAEPE

Substituting Equation 4 into Equation 3 yields

$$F(\theta, \varphi) = g^{\text{ave}}(\theta, \varphi) \sum_{m=0}^{M_x-1} \sum_{n=0}^{M_y-1} \sum_{p=-\frac{P_x}{2}}^{\frac{P_x}{2}} \sum_{q=-\frac{P_y}{2}}^{\frac{P_y}{2}} c_{p,q}^{(m,n)} w_{m,n} e^{j\beta[(m+p)d_x u + (n+q)d_y v]} \quad (13)$$

As can be seen, the above expression is a kind of summation of complex exponentials. However, this expression has four-layer summation, and cannot be directly computed by applying the standard 2D-FFT. To solve this issue, we introduce a blocked accumulating matrix which relates the excitation vector of the enlarged virtual planar array and that of the real array. Note that since all the AEPs should be approximated by their corresponding virtual subarrays, the whole virtual array is a kind of overlapped combination of all the virtual subarrays, and naturally the same virtual element usually appear in different subarrays. Hence, accumulating multiple coupling coefficients from different virtual subarrays is essential for reducing the number of summations in Equation 13. The introduced blocked accumulating matrix provides a feasible way, and it is given by

$$\mathbf{A} = \begin{bmatrix} \mathbf{C}\left(0, -\frac{P_y}{2}\right) & & & \mathbf{O} \\ \vdots & \mathbf{C}\left(1, -\frac{P_y}{2}\right) & & \\ \mathbf{C}\left(0, \frac{P_y}{2}\right) & \vdots & \ddots & \\ & \mathbf{C}\left(1, \frac{P_y}{2}\right) & \vdots & \mathbf{C}\left(M_y - 1, -\frac{P_y}{2}\right) \\ & & \ddots & \vdots \\ \mathbf{O} & & & \mathbf{C}\left(M_y - 1, \frac{P_y}{2}\right) \end{bmatrix} \quad (14)$$

where $\mathbf{A} \in C^{[(M_x+P_x) \cdot (M_y+P_y)] \times M_x M_y}$ is a sparse matrix, and it can be divided into $(M_y + P_y) \times M_y$ blocked matrices. Among them, each non-zero blocked coupling matrix $\mathbf{C}(n, q) \in C^{(M_x+P_x) \times M_x}$ is also a sparse matrix with $n = 0, 1, \dots, M_y - 1$ and $q = -P_y/2, \dots, P_y/2$. The (l_x, m) th element of the matrix $\mathbf{C}(n, q)$ is given by

$$[\mathbf{C}(n, q)]_{l_x, m} = \begin{cases} c_{p,q}^{(m,n)}, & \text{for } l_x = m + p + \frac{P_x}{2} \text{ \& } |p| \leq \frac{P_x}{2} \\ 0, & \text{others.} \end{cases} \quad (15)$$

That is, all the coupling coefficients corresponding to the real array elements in the n th row are distributed in a total of $P_y + 1$ rows, and the coupling coefficients of each row can be assembled into a coupling matrix $\mathbf{C}(n, q)$. In each coupling matrix $\mathbf{C}(n, q)$, the different coupling coefficients from the same subarray are arranged in the same column, while multiple coupling coefficients of the same virtual element appearing in different subarrays are arranged in the same row. According to the arrangement of the above-mentioned blocked accumulating matrix, the real excitation vector can be defined as

$$\tilde{\mathbf{w}} = \text{vec} \left\{ \left[\mathbf{w}^{(0)}, \mathbf{w}^{(1)}, \dots, \mathbf{w}^{(M_y-1)} \right] \right\} \quad (16)$$

$$\mathbf{w}^{(n)} = [w_{0,n}, w_{1,n}, \dots, w_{M_x-1,n}]^T \quad (17)$$

where the $\text{vec}\{\cdot\}$ is the vectorization operator of a matrix. Then, the virtual excitation vector can be obtained by multiplying the blocked accumulating matrix $\mathbf{A}$ and the real excitation vector $\tilde{\mathbf{w}}$, that is,

$$\tilde{\mathbf{s}} = \mathbf{A} \tilde{\mathbf{w}} \quad (18)$$

where

$$\tilde{\mathbf{s}} = \text{vec} \left\{ \left[\mathbf{s}^{(0)}, \mathbf{s}^{(1)}, \dots, \mathbf{s}^{(M_y+P_y-1)} \right] \right\} \quad (19)$$

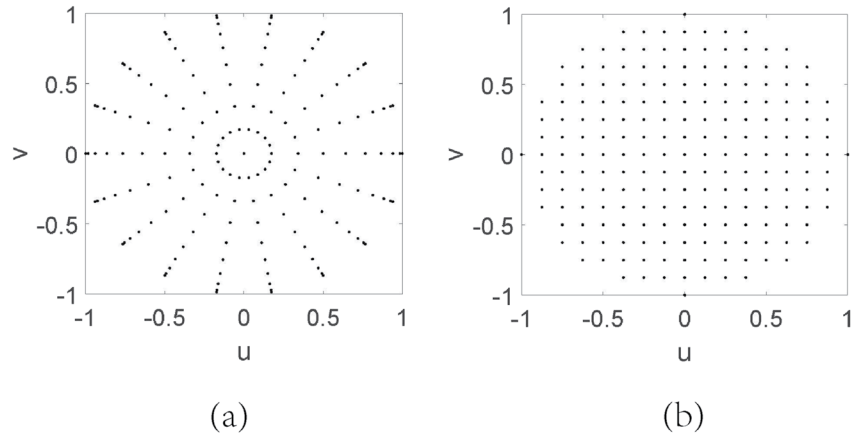


Figure 2. (a) Location of the spherical grid points (θ, ϕ) on the (u, v) -space. (b) Rectangular grid points on the (u, v) -space that is required for performing 2D FFT.

$$\mathbf{s}^{(l_y)} = [s_{0,l_y}, s_{1,l_y}, \dots, s_{M_x+P_x-1,l_y}]^T \quad (20)$$

In the above, s_{l_x,l_y} denotes the excitation of the (l_x, l_y) th element in the enlarged virtual planar array where $l_x = 0, 1, \dots, M_x + P_x - 1$ and $l_y = 0, 1, \dots, M_y + P_y - 1$. Now, with the relationship between the virtual excitation vector and the real excitation vector in Equation 18 and Equation 13 can be reduced to a two-layer summation that is given by

$$F(\theta, \varphi) = g^{\text{ave}}(\theta, \varphi) e^{-j\beta \left(\frac{P_x}{2} d_x u + \frac{P_y}{2} d_y v \right)} \sum_{l_x=0}^{L_x-1} \sum_{l_y=0}^{L_y-1} s_{l_x,l_y} e^{j\beta(l_x d_x u + l_y d_y v)} \quad (21)$$

where $L_x = M_x + P_x$ and $L_y = M_y + P_y$ represent the element number of the enlarged virtual planar array along the x -axis and the y -axis, respectively. Obviously, the expression of Equation 21 indicates a FT-pair relationship exists between the virtual excitations and the array pattern, and thus the pattern computation can be speeded up by employing the 2D-FFT.

However, before using 2D-FFT to calculate Equation 21, one more thing to note is that the virtual element pattern $g^{\text{ave}}(\theta, \varphi)$ is obtained by averaging over all the 2D-AEPs. In practice, the 2D-AEPs are generally obtained by full-wave electromagnetic simulation or measurement, and the obtained element patterns are usually uniformly sampled over spherical angles θ and φ . For the spherical angles θ and φ with uniform sampling interval, according to the direction cosines $u = \sin\theta\cos\varphi$ and $v = \sin\theta\sin\varphi$, the corresponding grid points on the (u, v) -space is calculated, as shown in as shown in Figure 2a. However, such pattern sampling grid in Figure 2a does not conform with the 2D-FFT requirement, since performing the 2D-FFT requires a rectangular sampling grid in the (u, v) -space, such as Figure 2b. In order to solve this issue, an interpolation-based pre-processing technique is presented to obtain the samples of the virtual element pattern in the uniform grid of (u, v) -space. Here, the interpolation is implemented via MATLAB's built-in function of *scatteredInterpolant* which uses Delaunay triangulation of the scattered sampling points (Amidror, 2002). In this way, the virtual element pattern $g^{\text{ave}}(u, v)$ in the (u, v) -space is obtained. And then, Equation 21 can be rewritten as

$$F(u, v) = g^{\text{ave}}(u, v) e^{-j\beta \left(\frac{P_x}{2} d_x u + \frac{P_y}{2} d_y v \right)} \sum_{l_x=0}^{L_x-1} \sum_{l_y=0}^{L_y-1} s_{l_x,l_y} e^{j\beta(l_x d_x u + l_y d_y v)} \quad (22)$$

When u and v are discretized by

$$\begin{aligned} u &= k_u \Delta u = k_u \frac{\lambda}{K_u d_x} \\ v &= k_v \Delta v = k_v \frac{\lambda}{K_v d_y} \end{aligned} \quad (23)$$

where $k_u = -K_u/2, \dots, K_u/2 - 1$ ($K_u \geq L_x$) and $k_v = -K_v/2, \dots, K_v/2 - 1$ ($K_v \geq L_y$), Equation 22 becomes

Algorithm 1. The Proposed Iterative 2D-LSAEPE-FFT for Pattern Synthesis of Planar Arrays With Radomes

1. According to the user-defined desired polarization direction $\vec{p}_d$, compute the component of each 2D-AEP in the desired direction $g^{(m,n)}(\theta, \varphi)$.
2. Give the desired pattern requirements: set the upper bound and the lower bound of array patterns.
3. Choose suitable parameters P_x and P_y , and then compute the coupling coefficient vector $\tilde{\mathbf{c}}^{(m,n)}$ and the blocked accumulating matrix $\mathbf{A}$.
4. Perform an interpolation-based pre-processing technique to obtain the virtual element pattern $g^{\text{ave}}(u, v)$ in uniform grid of the (u, v) -space.
5. Initialize the excitation for the original array $\tilde{\mathbf{w}}$.
6. Compute the excitation for the enlarged virtual planar array by $\tilde{\mathbf{s}} = \mathbf{A}\tilde{\mathbf{w}}$.
7. Calculate the array pattern $F(k_u, k_v)$ through performing a $K_u \times K_v$ points 2D-IFFT on the zero-padded $\tilde{\mathbf{s}}$ in Equation 24.
8. Check if the obtained pattern $F(k_u, k_v)$ satisfy the prescribed boundary. If yes, stop the iterative procedure; otherwise, adjust the magnitudes of $F(k_u, k_v)$ to adapt to the bounds while keeping the phases unchanged.
9. Calculate the updated excitation $\tilde{\mathbf{s}}$ from the modified $F(k_u, k_v)$ through performing the $K_u \times K_v$ points 2D-FFT in Equation 25. Only $L_x \times L_y$ points are remained.
10. Compute the real excitation $\tilde{\mathbf{w}} = (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \tilde{\mathbf{s}}$.
11. Repeat Step 6 to 10 before all the requirements are satisfied or reaching the maximum allowable number of iterations.

$$F(k_u, k_v) = g^{\text{ave}}(k_u, k_v) e^{-j\pi \left(\frac{P_x}{K_u} k_u + \frac{P_y}{K_v} k_v \right)} \sum_{l_x=0}^{L_x-1} \sum_{l_y=0}^{L_y-1} s_{l_x, l_y} e^{j \frac{2\pi}{K_u} k_u l_x} e^{j \frac{2\pi}{K_v} k_v l_y} \quad (24)$$

Clearly, this summation can be calculated efficiently by performing the $K_u \times K_v$ -point 2D-IFFT after extending the initial sequence $\{s_{l_x, l_y}; l_x = 0, 1, \dots, L_x - 1 \text{ and } l_y = 0, 1, \dots, L_y - 1\}$ to be $K_u \times K_v$ -length through the zero-padding technique. In such way, the obtained pattern includes the mutual coupling and radome effect.

Conversely, if the array pattern of the $L_x \times L_y$ enlarged virtual planar array is known, the virtual excitation vector can also be efficiently obtained. From Equation 24, its inverse transform can be expressed as

$$s_{l_x, l_y} = \sum_{k_u=-\frac{K_u}{2}}^{\frac{K_u}{2}-1} \sum_{k_v=-\frac{K_v}{2}}^{\frac{K_v}{2}-1} \frac{F(k_u, k_v)}{g^{\text{ave}}(k_u, k_v) e^{-j\pi \left(\frac{P_x}{K_u} k_u + \frac{P_y}{K_v} k_v \right)}} e^{-j \frac{2\pi}{K_u} k_u l_x} e^{-j \frac{2\pi}{K_v} k_v l_y} \quad (25)$$

This summation in Equation 25 can obviously be solved quickly by the $K_u \times K_v$ -point 2D-FFT. Once the virtual excitation vector $\tilde{\mathbf{s}}$ is obtained, the real excitation vector $\tilde{\mathbf{w}}$ can be calculated by the least squares solution of Equation 18. That is, $\tilde{\mathbf{w}} = (\mathbf{A}^H \mathbf{A})^{-1} \mathbf{A}^H \tilde{\mathbf{s}}$.

2.4. The Proposed Iterative 2D-LSAEPE-FFT for Accurate Pattern Synthesis of Planar Arrays With Radomes

The detailed process of the proposed iterative 2D-LSAEPE-FFT for accurate pattern synthesis of planar arrays with radomes is listed in Algorithm 1. In comparison with the traditional IFT (Keizer, 2007), the proposed synthesis procedure integrates the 2D-LSAEPE method into IFT, which can achieve accurate pattern synthesis of planar arrays considering mutual coupling and radome effect, while maintaining the computational efficiency of the FFT. It is assumed that all the 2D-AEPs have been obtained by full-wave simulating or measuring the whole array-radome structure before the proposed iterative pattern synthesis procedure.

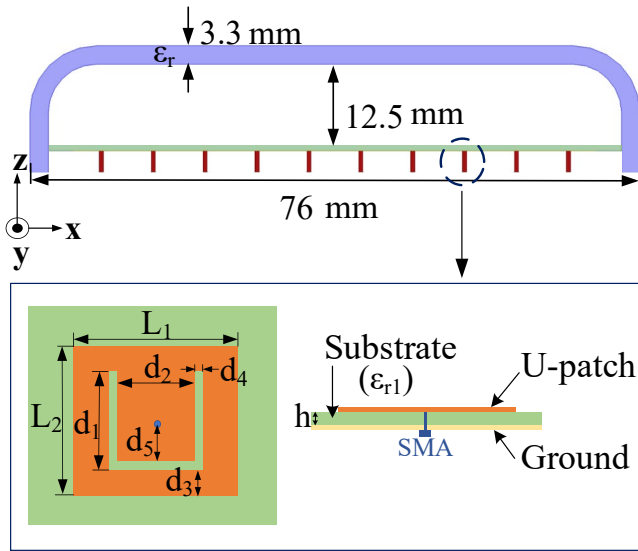


Figure 3. Geometry of a 10×10 -element planar array with a dielectric radome. The parameters of the U-slot microstrip antenna are: $L_1 = 4.5$ mm, $L_2 = 3.6$ mm, $d_1 = 2.3$ mm, $d_2 = 1.76$ mm, $d_3 = 0.4$ mm, $d_4 = 0.2$ mm, $d_5 = 1.2$ mm, $h = 0.79$ mm, and $\epsilon_{r1} = 2.2$.

3. Numerical Results

In this section, several examples for synthesizing radome-enclosed planar arrays with different pattern requirements are provided to verify the accuracy and efficiency of the proposed method. In each example, all the 2D-AEPs used in the pattern synthesis procedure are obtained by performing one-time full-wave simulation. The statistical time consumption in each example does not include the full-wave simulation process. All the synthesis examples are executed on the same computer with Intel Core i5-10400F CPU@2.90 GHz.

3.1. Pencil Beam Synthesis of a 10×10 U-Slot Microstrip Antenna Array Enclosed With a Flat Radome

In the first example, we consider to synthesize a pencil beam for a 10×10 planar array placed inside a flat single-layer dielectric radome. The array element uses a U-slot microstrip antenna with a resonant frequency of 24 GHz, and the detailed design procedure of this kind of antenna can be found in Weigand et al. (2003). The spacing of adjacent elements is 6.875 mm along both the x -axis and y -axis (i.e., 0.55λ at 24 GHz). The geometry of this array-radome system is shown in Figure 3. The radome is made of polybutylece terephthalate (PBT) with relative permittivity of $\epsilon_r = 3.5$, loss tangent $\tan\delta = 0.0027$ and thickness of $d = 3.3$ mm. The distance of the array from the radome is 12.5 mm. Suppose the desired polarization direction is $\hat{y}$ and the maximum SLL is required to be lower than -25 dB for the sidelobe

region of $|\theta| \geq 13.75^\circ$ and $\varphi \in (0^\circ, 360^\circ)$. If we directly apply the traditional IFT algorithm which handles only the array factor to synthesize this planar array without/with radome, the obtained real array pattern (φ -cuts in step of 2°) by using full-wave simulated AEPs with excitations derived from the traditional IFT are plotted in Figures 4a and 4b, respectively. As can be seen, the presence of the radome causes a great deterioration of the array pattern in the sidelobe region, and the maximum SLL rises from -25 dB to -20.2 dB. This illustrates that the traditional IFT without considering radome effect into the pattern synthesis is not applicable.

Now, we apply the proposed iterative 2D-LSAEPE-FFT to synthesize the same example. The 2D-AEP for each element is approximated as a pattern radiated by a virtual planar sub-array consisting of nearby elements. The approximation error can be defined as

$$\epsilon = \frac{\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \|\mathbf{g}^{(m,n)} - \mathbf{Z}\tilde{\mathbf{c}}^{(m,n)}\|_2^2}{\sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \|\mathbf{g}^{(m,n)}\|_2^2}. \quad (26)$$

In this example, we choose P_{xy} in Equation 4 as 6 for obtaining $\epsilon \leq 1\%$ which gives a good balance between synthesis accuracy and efficiency. Figure 5 shows two orthogonal φ -cuts of the real and approximated 2D-AEPs of some typical elements when with or without a radome. It can be seen that the presence of the radome does change the shapes of 2D-AEPs considerably, and it affects the edge element more than the inner element. For both cases of the array with/without the radome, the approximated AEPs by using the proposed 2D-LSAEPE agree well with the real AEPs. The synthesized array pattern and the corresponding real array pattern for the case with radome by the proposed method (the synthesized and real array patterns are obtained by using the synthesized excitations with the approximated and real AEPs, respectively) are shown in Figures 4c and 4d, respectively. As can be seen, both the synthesized and real array patterns completely satisfy the prescribed bounds. Figure 6 shows the excitation amplitude and phase distributions obtained by the traditional IFT and the proposed iterative 2D-LSAEPE-FFT. In this example, the time cost for the traditional IFT and the proposed method are 0.28 and 2.17 s, respectively.

3.2. Shaped Pattern Synthesis of a 12×12 Printed Bowtie Antenna Array Enclosed With a Curved Radome

In the second example, to further verify the robustness of the proposed iterative 2D-LSAEPE-FFT method for different pattern shape requirements encountered in practical applications, we consider to synthesize a

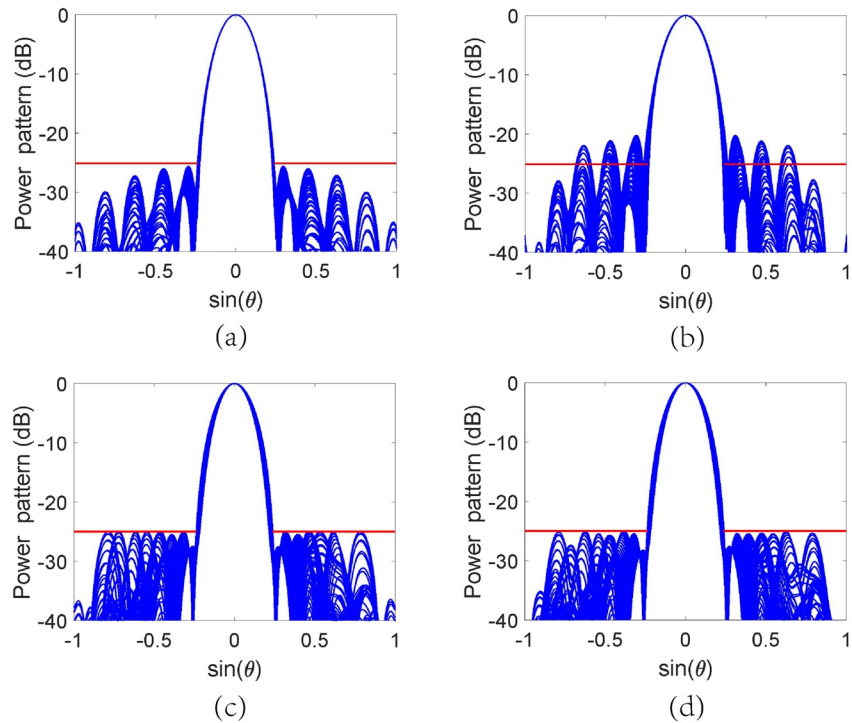


Figure 4. φ -cuts (in step of 2°) of the obtained array patterns. Panels (a and b) show the real pattern without/with the radome using the traditional IFT in Keizer (2007), respectively. Panels (c and d) show the synthesized and real patterns with the radome using the proposed method, respectively.

crescent-shaped pattern. The array element adopts a printed bowtie antenna with a center frequency of 5.8 GHz, whose geometry is shown in Figure 7. The interelement spacing is 15.275 mm along both the x -axis and y -axis (i.e., 0.5λ at 5.8 GHz) in the planar array. The radome used in this section is a curved structure with the geometry shown in Figure 7. Its material is the same as that of Example 1, but the thickness becomes 13.8 mm due to the change of operating frequency. For the pattern synthesis in this section, the $P_{x/y}$ is chosen as 8 for $\epsilon \leq 1\%$, and the initial excitations are obtained by using Woodward-Lawson method (Woodward & Lawson, 1948).

Assume that the desired pattern has a crescent-shaped mainlobe with $-2 \text{ dB} \leq |F(u, v)|^2 \leq 0 \text{ dB}$ in the region of $\{(u, v); |u^2 + v^2 \leq 0.35^2 \text{ and } (u + 0.35)^2 + v^2 \geq 0.5^2\}$, and the SLL is less than -25 dB in the region of $\{(u, v); |u + 0.5|^2 + v^2 \leq 0.5^2 \text{ or } u^2 + v^2 \geq 0.5^2\}$. Figures 8a and 8b show top views of the synthesized and real patterns by the proposed method, respectively. As can be seen, they are almost consistent, and both of them satisfy the pre-specified constraint. For comparison, the SDR method in (Y. Liu et al., 2018) and the PSO method in Pirhadi et al. (2014) are also adopted to synthesize this crescent-shaped pattern. Figures 8c and 8d show the real array patterns obtained by these two methods, respectively. The synthesized result by the SDR method is quite satisfactory since it directly uses accurate 2D-AEPs in the synthesis process. For the PSO method, the number of iterations and the size of the population are set as 2,000 and 288, respectively. The best pattern result obtained over 20 runs is selected, and its maximum SLL is slightly higher than that obtained by the proposed method. In this example, the SDR and PSO methods take 2928.51 and 2475.42 s, respectively, while the proposed method takes only 6.89 s.

3.3. Scannable Beam Synthesis of a 8×8 Printed Dipole Antenna Array Enclosed With a Superspheroidal Radome

In the last example, a 8×8 planar array placed inside a radome with superspheroidal shape is considered. This kind of geometry is very popular in radome designs for high-speed vehicles as it preserves the vehicle's aerodynamic properties. In Overfelt (1995), Overfelt provided a unified description of a family of superspheroidal radome shapes. The surface of a superspheroidal body is given by $y^2 + z^2 = (D/2L)^2 (L^v - x^v)^{2/v}$, where D is the base diameter, L is the length of a radome, and v is any positive real number. Different v can generate different

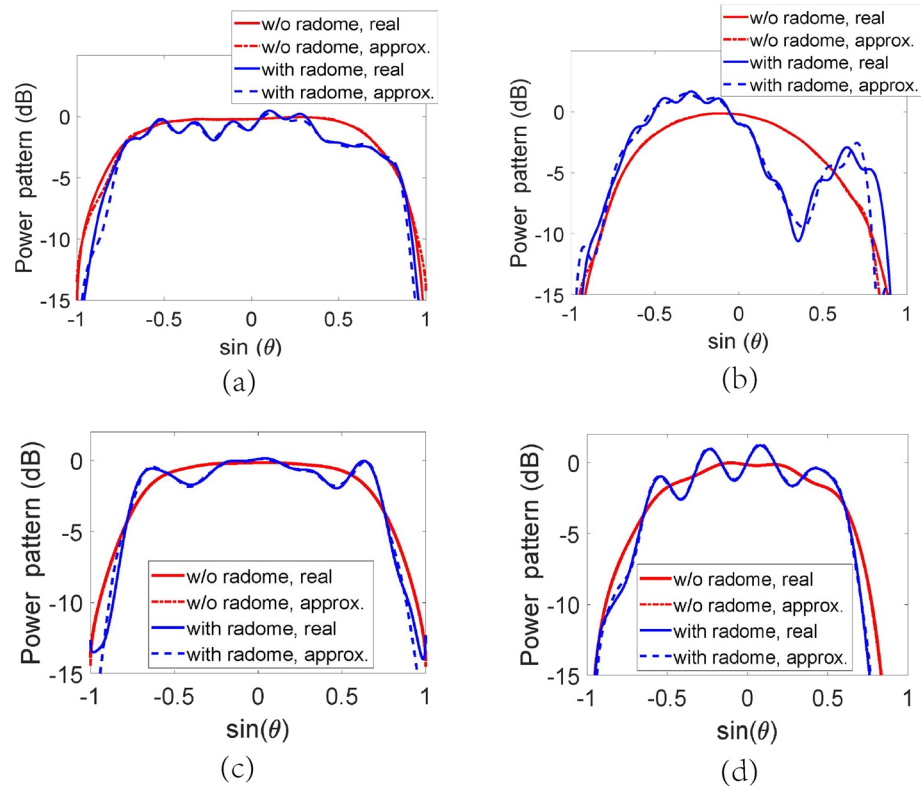


Figure 5. Panels (a and b) are the $\varphi = 0^\circ$ and $\varphi = 90^\circ$ cuts of the real active element patterns (AEPs) (solid line) and the approximated ones (dashed line) with/without radome for the element at the edge ($m = n = 1$), respectively; (c and d) are the AEPs for the middle element ($m = n = 5$) at $\varphi = 0^\circ$ and 90° , respectively.

radome shapes such as Sears-Haack ($\nu = 1.555$), Von Karman ($\nu = 1.381$), tangent ogive ($\nu = 1.449$), and 3/4 Power Law ($\nu = 1.161$). In this example, a Sears-Haack radome is adopted and its geometry is shown in Figure 9a. The element uses a printed dipole antenna resonating at 10 GHz, whose structure is plotted in Figure 9b. The material of substrate1 and substrate2 used in this antenna are the F4B with relative permittivity of $\epsilon_r = 4.4$ and thickness of $d = 0.5$ mm, and the FR4 with relative permittivity of 4.4 and thickness of 1 mm, respectively. The spacing of adjacent elements is 16.5 mm along both the x -axis and y -axis (i.e., 0.55λ at 10 GHz) for mounting supports in manufacture. It is assumed that the desired array pattern can be scanned from $\theta_0 = -30^\circ$ – 30° for any φ_0 , that is, $\theta_0 \in [-30^\circ, 30^\circ]$ and $\varphi_0 \in [0^\circ, 360^\circ]$. The maximum SLL is expected to be lower than -20 dB over the sidelobe region of $\{(u, v); |(u - u_0)^2 + (v - v_0)^2 \geq 0.27^2\}$ ($u_0 = \sin \theta_0 \cos \varphi_0$ and $v_0 = \sin \theta_0 \sin \varphi_0$). Figures 10a and 10b show the top views of the synthesized array patterns pointing at broadside and scanned to direction of $(\theta_0, \varphi_0) = (30^\circ, 30^\circ)$ by the proposed method, respectively. The corresponding real array patterns obtained by using full-wave simulation are shown in Figures 10c and 10d. It can be seen that the synthesized array pattern agrees well with that of the full-wave simulated result, regardless of whether the beam is scanned or not.

Then, the prototype of this array-radome structure is fabricated, as shown in Figure 11a. The radome shown is a 3D printed model made of resin. This antenna array-radome prototype is measured using a spherical near-field test system in a microwave anechoic chamber as shown in Figure 11b. In the test, only one antenna element is fed and all the others are connected to matched loads to obtain the measured AEPs of that excited element. In this way, the measured AEPs of all 64 elements can be obtained successively. Through using measured AEPs with the excitation amplitudes and phases obtained by the proposed method, the measured array patterns are acquired, which are drawn in Figures 10e and 10f for comparison. As can be seen in Figure 10, the measured maximum SLLs are -18.75 and -17.67 dB for the broadside directed beam and the beam scanned to direction of $(30^\circ, 30^\circ)$, respectively. Although the pattern performance may be slightly degraded due to some fabrication errors and non-ideal measurement environment, the measured array patterns in total agree well with the simulated results. It shows that by optimizing the excitation amplitude and phase, the influence of the mutual coupling between array

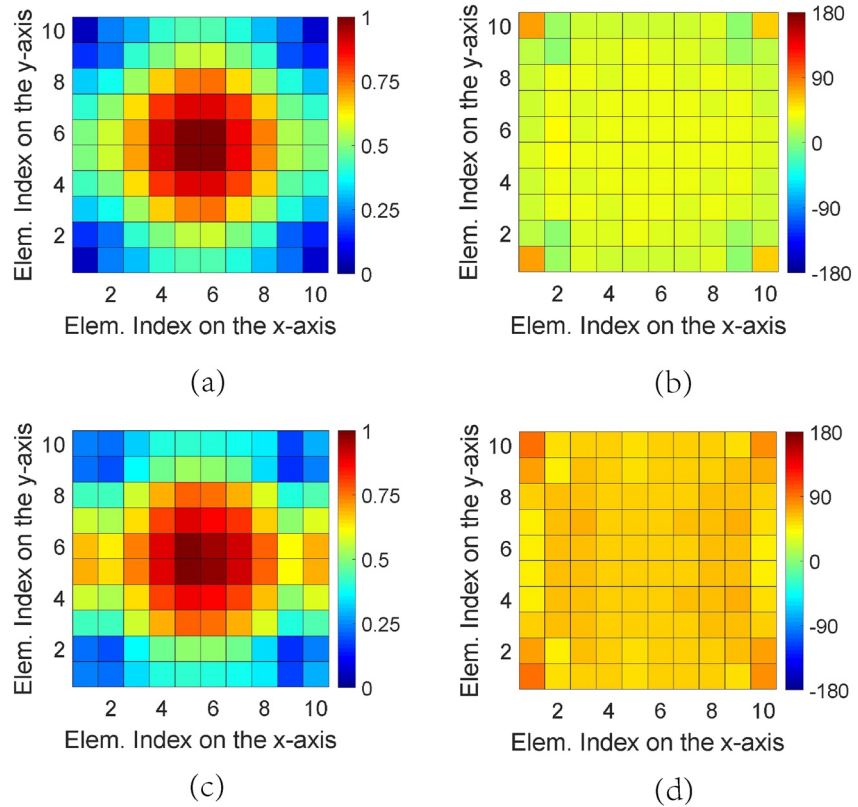


Figure 6. Excitation distributions obtained by the traditional IFT and the proposed iterative 2D-LSAEPE-FFT. Panels (a and b) show the amplitudes and phases by the traditional IFT, respectively, and (c and d) show the ones by the proposed iterative 2D-LSAEPE-FFT, respectively.

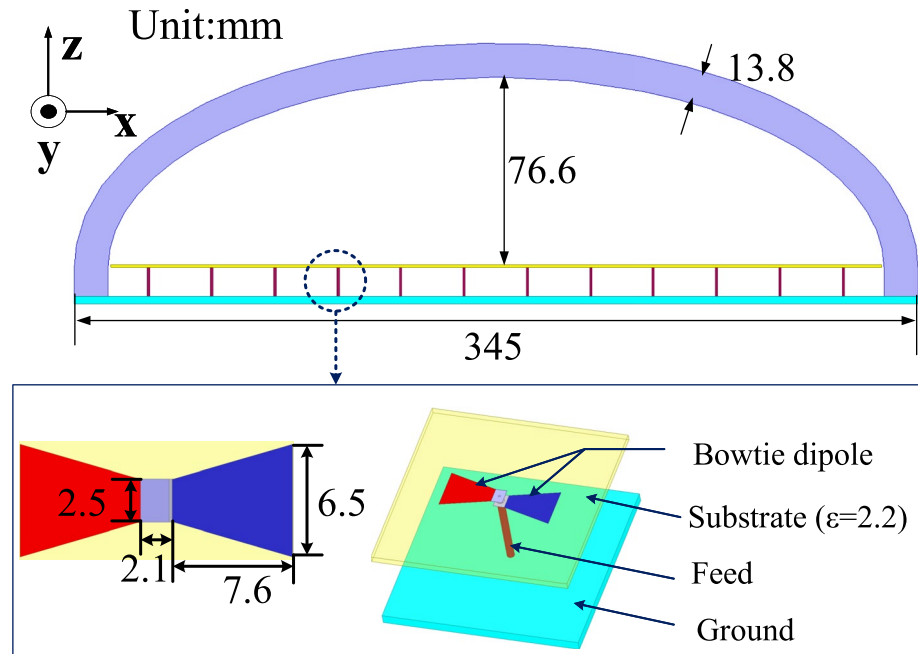


Figure 7. Geometry of a 12×12 -element planar array with printed bowtie antennas enclosed with a curved radome.

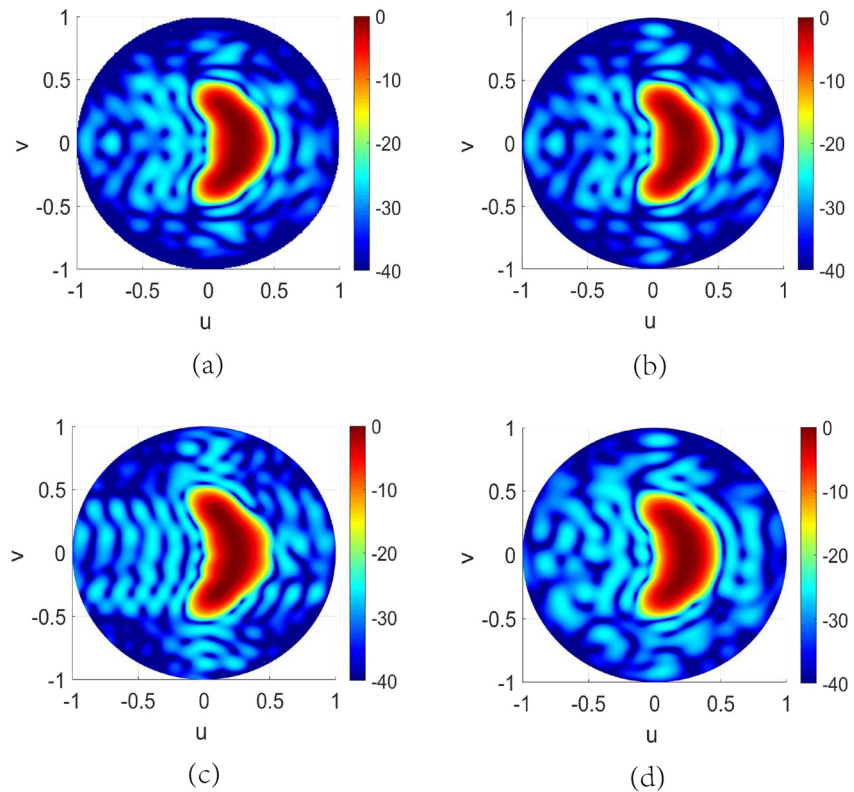


Figure 8. Top views of the obtained results by different methods. Panels (a and b) are the synthesized and real patterns by the proposed iterative 2D-LSAEPE-FFT, respectively. Panels (c and d) are the real patterns by the semidefinite relaxation method (Y. Liu et al., 2018) and the particle swarm optimization method (Pirhadi et al., 2014), respectively.

elements and the existence of radome on the distortion of the array pattern is well suppressed. This verifies the effectiveness of the proposed method for a practical array-radome system.

4. Conclusion

We have presented an efficient and accurate pattern synthesis method for radome-enclosed planar arrays in radar applications. The 2D-AEPs are adopted to include the mutual coupling and the influence of radome. The

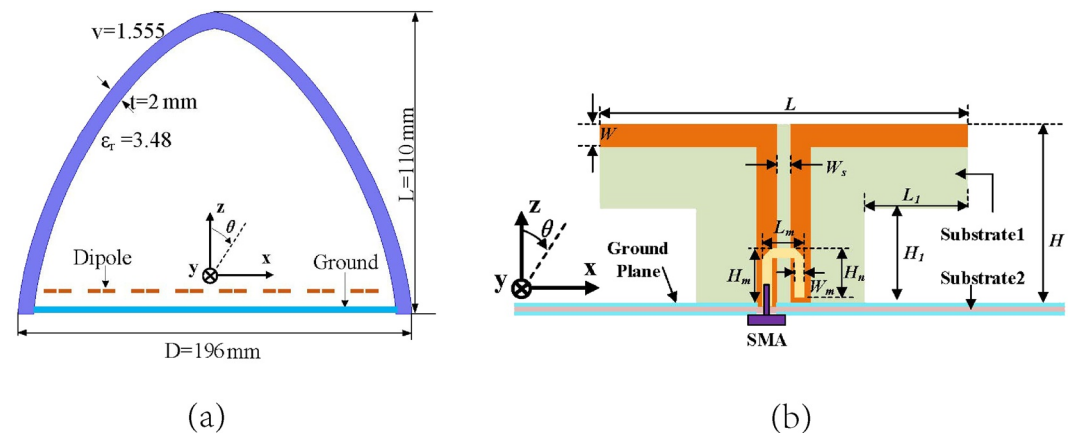


Figure 9. Geometries of an array-radome system. (a) Geometrical parameters of the used Sears-Haack radome. (b) Geometry of the printed dipole. The parameters are: $L = 14$; $H = 8$; $W = 1$; $W_s = 1$; $L_1 = 3$; $H_1 = 4.7$; $L_m = 2.62$; $H_m = 3.32$; $H_n = 2.18$; $W_m = 0.62$. (Unit:mm).

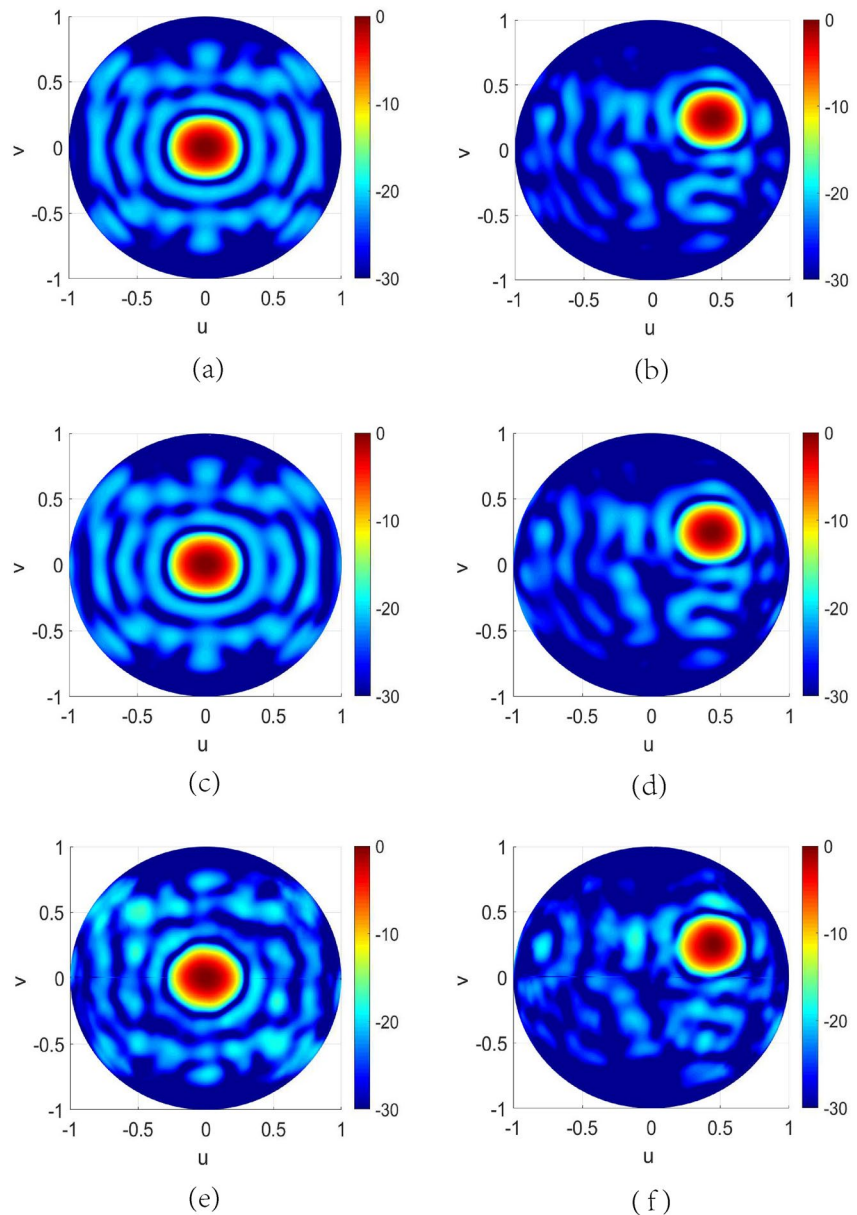


Figure 10. Top views of array patterns pointing at broadside and scanned to direction of $(\theta_0, \varphi_0) = (30^\circ, 30^\circ)$, respectively. Panels (a and b) the synthesized patterns; (c and d) the full-wave simulated patterns; (e and f) the measured patterns.

2D-LSAEPE is developed in which each AEP is regarded as the pattern radiated by a nearby virtual rectangular subarray. To integrate the FFT with 2D-LSAEPE for planar array case, a blocked accumulating matrix is introduced which relates the virtual and physical antenna array excitation vectors. In addition, an interpolation-based pre-processing technique is presented to obtain the samples of the virtual element pattern in the uniform grid of (u, v) -space. By integrating the FFT with the proposed 2D-LSAEPE as well as the interpolation technique, an iterative method is developed to achieve accurate and efficient pattern synthesis for planar arrays enclosed with radomes. Three synthesis experiments are carried out: the first is for synthesizing 10×10 U-slot microstrip antenna array enclosed with a flat radome, the second is for optimizing 12×12 printed bowtie antenna array with a curved radome, and the third is for considering a 8×8 printed dipole antenna array with a superspheroidal radome. Synthesis results show that the proposed method achieves much more accurate synthesis results compared with the traditional IFT technique. For the first numerical example, the obtained maximum SLL are -20.2 and -25 dB by the traditional IFT and the proposed method, respectively. Additionally, the proposed method takes much less computation time than both PSO and SDR methods, when the performance of the synthesized results

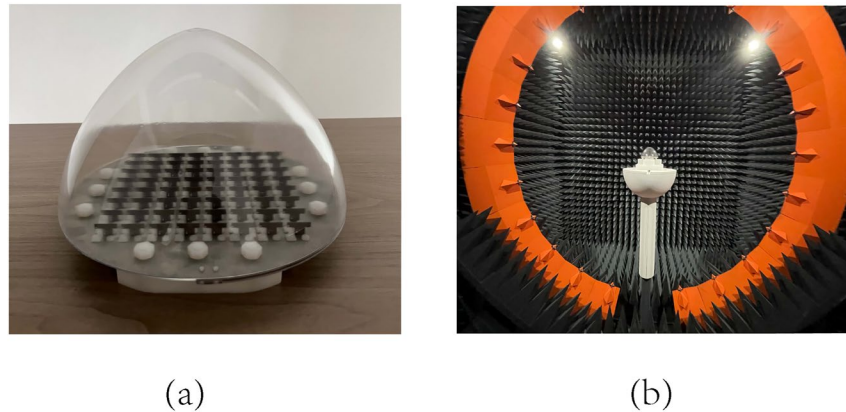


Figure 11. (a) The manufactured 8×8 printed dipole antenna array together with a superspheroidal radome. (b) The photo of this array-radome structure under test in a microwave anechoic chamber.

is comparable. For the test examples, the proposed method only costs no more than 7 s while both PSO and SDR methods take more than 2,000 s (all synthesis examples are performed on the same personal computer with Intel Core i5-10400F CPU@2.90 GHz). A prototype of 8×8 printed dipole antenna array-radome structure is fabricated, and the measured results are in good agreement with the simulated results.

Data Availability Statement

Numerical data related to Section 3.1 of this paper is available from the general repository Zenodo (<https://doi.org/10.5281/zenodo.7318022>).

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References

- Alibakhshikenari, M., Ali, E. M., Soruri, M., Dalarsson, M., Naser-Moghadasi, M., Virdee, B. S., et al. (2022). A comprehensive survey on antennas on-chip based on metamaterial, metasurface, and substrate integrated waveguide principles for millimeter-waves and terahertz integrated circuits and systems. *IEEE Access*, 10, 3668–3692. <https://doi.org/10.1109/ACCESS.2021.3140156>
- Alibakhshikenari, M., Virdee, B. S., Althuwayb, A. A., Aïssa, S., See, C. H., Abd-Alhameed, R. A., et al. (2021). Study on on-chip antenna design based on metamaterial-inspired and substrate-integrated waveguide properties for millimetre-wave and THz integrated-circuit applications. *Journal of Infrared, Millimeter and Terahertz Waves*, 42(1), 17–28. <https://doi.org/10.1007/s10762-020-00753-8>
- Alibakhshikenari, M., Virdee, B. S., Benetatos, H., Ali, E. M., Soruri, M., Dalarsson, M., et al. (2022). An innovative antenna array with high inter element isolation for sub-6 GHz 5G MIMO communication systems. *Scientific Reports*, 12(1), 7907. <https://doi.org/10.1038/s41598-022-12119-2>
- Alibakhshikenari, M., Virdee, B. S., Salekzamankhani, S., Aïssa, S., See, C. H., Soin, N., et al. (2021). High-isolation antenna array using SIW and realized with a graphene layer for sub-terahertz wireless applications. *Scientific Reports*, 11(1), 1–14. <https://doi.org/10.1038/s41598-021-87712-y>
- Alibakhshikenari, M., Virdee, B. S., Vadalà, V., Dalarsson, M., de Cos Gómez, M. E., Alharbi, A. G., et al. (2022). Broadband 3-D shared aperture high isolation nine-element antenna array for on-demand millimeter-wave 5G applications. *Optik*, 267, 169708. <https://doi.org/10.1016/j.ijleo.2022.169708>
- Amidror, I. (2002). Scattered data interpolation methods for electronic imaging systems: A survey. *Journal of Electronic Imaging*, 11(2), 157–176. <https://doi.org/10.1117/1.1455013>
- Ares-Pena, F., Rodriguez-Gonzalez, J., Villanueva-Lopez, E., & Rengarajan, S. (1999). Genetic algorithms in the design and optimization of antenna array patterns. *IEEE Transactions on Antennas and Propagation*, 47(3), 506–510. <https://doi.org/10.1109/8.768786>
- Bai, J., Liu, Y., Ren, Y., Nie, Z., & Guo, Y. J. (2021). Efficient synthesis of linearly polarized shaped patterns using iterative FFT via vectorial least-square active element pattern expansion. *IEEE Transactions on Antennas and Propagation*, 69(9), 6040–6045. <https://doi.org/10.1109/TAP.2021.3061574>
- Capozzoli, A., Curcio, C., D’Elia, G., & Lisenio, A. (2022). A technique for the array partitioning. *Radio Science*, 57(10), 1–22. <https://doi.org/10.1029/2021RS007384>
- Caratelli, D., & Viganò, M. (2011). Analytical synthesis technique for linear uniform-amplitude sparse arrays. *Radio Science*, 46(4). <https://doi.org/10.1029/2010RS004522>
- Chou, H.-T., Hsaio, Y.-T., Pathak, N., & Janpugdee (2004). A fast DFT planar array synthesis tool for generating contoured beams. *IEEE Antennas and Wireless Propagation Letters*, 3, 287–290. <https://doi.org/10.1109/LAWP.2004.837504>
- Dai, L., Xie, Y.-J., Zhang, C., & Wu, P. (2021). Fast optimization of array antenna enclosed by asymmetric radome using AEP combined with enhanced HGAPSO. *Progress in Electromagnetics Research M*, 103, 161–171. <https://doi.org/10.2528/PIERM21051306>
- D’Urso, M., Prisco, G., & Tumolo, R. M. (2016). Maximally sparse, steerable, and nonsuperdirective array antennas via convex optimizations. *IEEE Transactions on Antennas and Propagation*, 64(9), 3840–3849. <https://doi.org/10.1109/TAP.2016.2586490>
- Fenn, A. J., Temme, D. H., Delaney, W. P., & Courtney, W. E. (2000). The development of phased-array radar technology. *Lincoln Laboratory Journal*, 12(2), 321–340.

- Frech, M., Lange, B., Mammen, T., Selmann, J., Morehead, C., & Rowan, J. (2013). Influence of a radome on antenna performance. *Journal of Atmospheric and Oceanic Technology*, 30(2), 313–324. <https://doi.org/10.1175/JTECH-D-12-00033.1>
- Frid, H., & Jonsson, B. L. G. (2018). Compensation of radome effects in small airborne monopulse arrays by convex optimization. In *Paper presented at 12th European Conference on Antennas and Propagation (EUCAP 2018)* (pp. 1–5). <https://doi.org/10.1049/cp.2018.0647>
- Fuchs, B. (2012). Synthesis of sparse arrays with focused or shaped beam pattern via sequential convex optimizations. *IEEE Transactions on Antennas and Propagation*, 60(7), 3499–3503. <https://doi.org/10.1109/TAP.2012.2196951>
- Hsu, F., Chang, P.-R., & Chan, K.-K. (1993). Optimization of two-dimensional radome boresight error performance using simulated annealing technique. *IEEE Transactions on Antennas and Propagation*, 41(9), 1195–1203. <https://doi.org/10.1109/8.247745>
- Huang, X., Liu, Y., You, P., Zhang, M., & Liu, Q. H. (2017). Fast linear array synthesis including coupling effects utilizing iterative FFT via least-squares active element pattern expansion. *IEEE Antennas and Wireless Propagation Letters*, 16, 804–807. <https://doi.org/10.1109/LAWP.2016.2604570>
- Keizer, W. P. M. N. (2007). Fast low-sidelobe synthesis for large planar array antennas utilizing successive fast Fourier transforms of the array factor. *IEEE Transactions on Antennas and Propagation*, 55(3), 715–722. <https://doi.org/10.1109/TAP.2007.891511>
- Keizer, W. P. M. N. (2009a). Large planar array thinning using iterative FFT techniques. *IEEE Transactions on Antennas and Propagation*, 57(10), 3359–3362. <https://doi.org/10.1109/TAP.2009.2029382>
- Keizer, W. P. M. N. (2009b). Low-sidelobe pattern synthesis using iterative Fourier techniques coded in MATLAB. *IEEE Antennas and Propagation Magazine*, 51(2), 137–150. <https://doi.org/10.1109/MAP.2009.5162038>
- Kelley, D., & Stutzman, W. (1993). Array antenna pattern modeling methods that include mutual coupling effects. *IEEE Transactions on Antennas and Propagation*, 41(12), 1625–1632. <https://doi.org/10.1109/8.273305>
- Koumou, G. (2010). *Radar technology*. BoD—Books on Demand.
- Landesa, L., Rodriguez, J., Obelleiro, F., & Pino, A. (1999). Pattern synthesis of array antennas in presence of dielectric bodies. *IEEE Transactions on Magnetics*, 35(3), 1522–1525. <https://doi.org/10.1109/20.767257>
- Li, W. T., Hei, Y. Q., Yang, J., & Shi, X. W. (2016). Synthesis of multiple-pattern planar arrays with a hybrid generalised iterative fast Fourier transform algorithm. *IET Microwaves, Antennas & Propagation*, 10(1), 16–24. <https://doi.org/10.1049/iet-map.2014.0595>
- Liu, L., & Nie, Z. (2019). Performance improvement of antenna array-radome system based on efficient compensation and optimization scheme. *IEEE Antennas and Wireless Propagation Letters*, 18(5), 866–870. <https://doi.org/10.1109/LAWP.2019.2903846>
- Liu, Y., Bai, J., Xu, K. D., Xu, Z., Han, F., Liu, Q. H., & Jay Guo, Y. (2018). Linearly polarized shaped power pattern synthesis with sidelobe and cross-polarization control by using semidefinite relaxation. *IEEE Transactions on Antennas and Propagation*, 66(6), 3207–3212. <https://doi.org/10.1109/TAP.2018.2816782>
- Overfelt, P. (1995). Superspheroids: A new family of radome shapes. *IEEE Transactions on Antennas and Propagation*, 43(2), 215–220. <https://doi.org/10.1109/8.366386>
- Pirhadi, A., Rahmani, M. H., & Mallahzadeh, A. (2014). Shaped beam array synthesis using particle swarm optimisation method with mutual coupling compensation and wideband feeding network. *IET Microwaves, Antennas & Propagation*, 8(8), 549–555. <https://doi.org/10.1049/iet-map.2013.0104>
- Poli, L., Rocca, P., Manica, L., & Massa, A. (2010). Handling sideband radiations in time-modulated arrays through particle swarm optimization. *IEEE Transactions on Antennas and Propagation*, 58(4), 1408–1411. <https://doi.org/10.1109/TAP.2010.2041165>
- Pozar, D. (1994). The active element pattern. *IEEE Transactions on Antennas and Propagation*, 42(8), 1176–1178. <https://doi.org/10.1109/8.310010>
- Wang, X.-K., Jiao, Y.-C., & Tan, Y.-Y. (2014). Synthesis of large thinned planar arrays using a modified iterative Fourier technique. *IEEE Transactions on Antennas and Propagation*, 62(4), 1564–1571. <https://doi.org/10.1109/TAP.2014.2302836>
- Weigand, S., Huff, G., Pan, K., & Bernhard, J. (2003). Analysis and design of broad-band single-layer rectangular U-slot microstrip patch antennas. *IEEE Transactions on Antennas and Propagation*, 51(3), 457–468. <https://doi.org/10.1109/TAP.2003.809836>
- Woodward, P., & Lawson, J. (1948). The theoretical precision with which an arbitrary radiation-pattern may be obtained from a source of finite size. *Journal of the Institution of Electrical Engineers-Part III: Radio and Communication Engineering*, 95(37), 363–370. <https://doi.org/10.1049/ji-3-2.1948.0094>
- Xu, W., Duan, B. Y., Li, P., & Qiu, Y. (2017). A new efficient thickness profile design method for streamlined airborne radomes. *IEEE Transactions on Antennas and Propagation*, 65(11), 6190–6195. <https://doi.org/10.1109/TAP.2017.2754460>
- Yang, K., Zhao, Z., & Liu, Q. H. (2013). Fast pencil beam pattern synthesis of large unequally spaced antenna arrays. *IEEE Transactions on Antennas and Propagation*, 61(2), 627–634. <https://doi.org/10.1109/TAP.2012.2220319>
- Zhou, P., Zhang, Z., & He, M. (2020). Radiation pattern recovery of the impaired-radome-enclosed antenna array. *IEEE Antennas and Wireless Propagation Letters*, 19(9), 1639–1643. <https://doi.org/10.1109/LAWP.2020.3013228>